Task 4

Do the exact thing as Task 2, but for a cylindrical coordinate system with axisymmetry. The diffusion equation then becomes:

$$\frac{\partial c}{\partial t} = D\frac{1}{r}\left(r\frac{\partial c}{\partial r}\right), \qquad 0 \le r \le 1$$

$$\text{or,} \quad \frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial r^2} + D\frac{1}{r}\frac{\partial c}{\partial r}, \qquad 0 \le r \le 1$$

Boundary conditions:

$$\text{At } r = 0, \quad \frac{\partial c}{\partial r} = 0,$$

$$\text{At } r = 1, \quad \frac{\partial c}{\partial r} = 1$$

Initial condition:

$$\text{At } t = 0, \quad c = 0.$$

Note that at $r = 0$, there is a bit of challenge with handling the governing equation because of the $\frac{1}{r}$ term. You can study how to handle it on page 251 of Langtangen's book, whose link I had shared earlier. Basically, you have to use the L'Hospital's rule.

Objective: Implementation of the numerical solution using finite difference method for cylindrical coordinate system with axissymmetry

Implementation: For the axisymmetric cylindrical diffusion equation, special attention was required at the boundary r = 0. The governing equation contains the term (1/r)(∂c/∂r), which becomes an indeterminate form (0/0) when the symmetry boundary condition ∂c/∂r = 0 is applied at r = 0. Therefore, L'Hospital's rule was used to evaluate the limit, resulting in a modified governing equation at the center node:

$\partial c/\partial t = 2D(\partial^2 c/\partial r^2)$

This expression was then used to derive the finite-difference equation at $r = 0$. The second boundary condition at $r = 1$,

$$\partial c/\partial r = 1$$

was implemented directly using a finite-difference approximation and a ghost node.

For the numerical implementation, the domain $0 \le r \le 1$ was discretized into $N = 50$ grid intervals. An explicit FTCS finite-difference scheme was employed with a Fourier number $Fo = 0.24$, ensuring numerical stability. The initial condition $c(r,0) = 0$ was applied throughout the domain, and the concentration profile was computed at successive time steps.

Calculations:

Task 4

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial r^2} + D\frac{1}{r}\frac{\partial c}{\partial r}$$

$0 \le r \le 1$

BC $\quad r=0 \quad \frac{\partial c}{\partial r} = 0$

$r=1 \quad \frac{\partial c}{\partial r} = 1$

$t=0 \quad c=0$

at $r=0$ $\quad \frac{\partial c}{\partial r} = 0$

$\therefore \frac{1}{r}\frac{\partial c}{\partial r} = \left(\frac{0}{0}\right)$ form

$\therefore$ applying L'hospitals rule,

$$\lim_{r\to 0}\frac{\frac{\partial c}{\partial r}}{r} = \lim_{r\to 0}\frac{\frac{\partial^2 c}{\partial r^2}}{1} = \left.\frac{\partial^2 c}{\partial r^2}\right|_{r=0}$$

$\therefore$ at $r=0$ $\quad \frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial r^2} + D\frac{\partial^2 c}{\partial r^2}$

$$\boxed{\frac{\partial c}{\partial t} = 2D\frac{\partial^2 c}{\partial r^2}}$$

grid formation

$r_i = i\,\Delta r$

if we form grid with N divisions

$i = 1, 2, 3, \ldots N$

$\Delta r = \frac{1}{N}$

Similarly, $t^n = n\Delta t$

$\therefore C_i^n = C(r_i, t^n)$

$\therefore$ now, for $i = 1, 2, \ldots N-1$

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} = D\left[\frac{C_{i+1}^n - 2C_i^n + C_{i-1}^n}{\Delta r^2} + \frac{1}{r_i}\frac{C_{i+1}^n - C_{i-1}^n}{2\Delta r}\right]$$

$$C_i^{n+1} - C_i^n = \frac{D\Delta t}{\Delta r^2}\left[C_{i+1}^n - 2C_i^n + C_{i-1}^n + \frac{\Delta r}{r_i}\frac{C_{i+1}^n - C_{i-1}^n}{2}\right]$$

let $F = \frac{D\Delta t}{\Delta r^2}$

$$C_i^{n+1} = C_i^n + F\left(C_{i+1}^n - 2C_i^n + C_{i-1}^n\right) + \frac{F\Delta r}{2r_i}\left(C_{i+1}^n - C_{i-1}^n\right)$$

at $r = 0$

$$\frac{\partial c}{\partial t} = 2D\frac{\partial^2 c}{\partial r^2}$$

$\Rightarrow \frac{C_i^{n+1} - C_i^n}{\Delta t} = 2D \left[\frac{C_{i+1}^n - 2C_i^n + C_{i-1}^n}{\Delta r^2} \right]$

$i = 0$

$\Rightarrow \frac{C_0^{n+1} - C_0^n}{\Delta t} = 2D \left[\frac{C_1^n - 2C_0^n + C_{-1}^n}{\Delta r^2} \right]$

$C_1 = C_{-1}$ (due to symmetry)

& $F = \frac{D \Delta t}{\Delta r^2}$

$\therefore C_0^{n+1} = C_0^n + 4F_0 (C_1^n - C_0^n)$

at $r = 1$ $\left. \frac{\partial c}{\partial r} \right|_{r=1} = 1$

$\Rightarrow \frac{C_{N+1} - C_{N-1}}{2\Delta r} = 1$

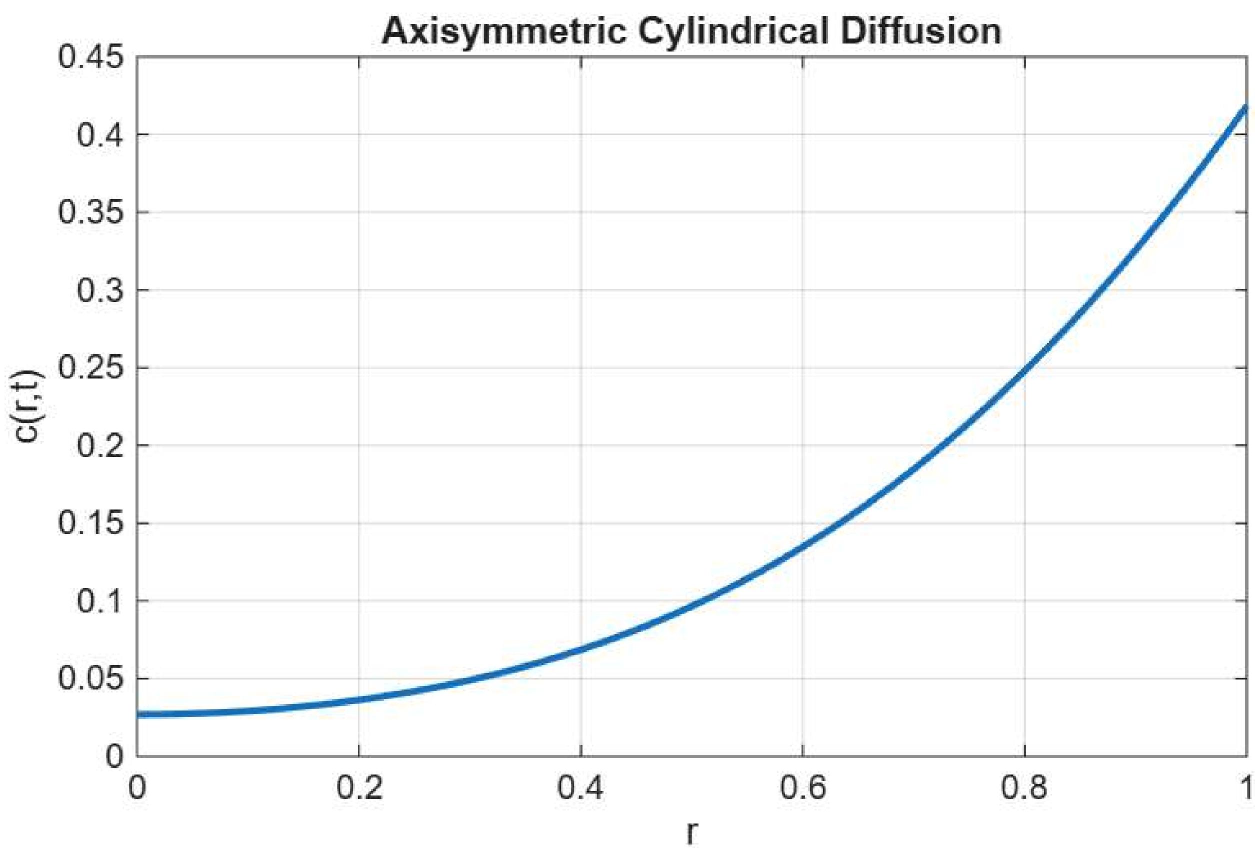
$\Rightarrow C_{N+1} = C_{N-1} + 2\Delta r$

& initially, $C_i^0 = 0 \quad \forall i \in \{0, N\}$

Matlab Implementation:

```
clc;
clear;
close all;
D = 1;
N = 50;
dr = 1/N;
Fo = 0.24;
dt = Fo*dr^2/D;
t_final = 0.1;
nt = round(t_final/dt);
r = linspace(0,1,N+1);
c = zeros(N+1,1);
for n = 1:nt
    c_old = c;
    % at r=0 (center node)
    c(1) = c_old(1) + 4*Fo*(c_old(2)-c_old(1));
    for i = 2:N
        ri = r(i);
        c(i) = c_old(i) ...
             + Fo*(c_old(i+1)-2*c_old(i)+c_old(i-1)) ...
             + (Fo*dr/(2*ri))*(c_old(i+1)-c_old(i-1));
    end
    % at r=1
    ghost = c_old(N) + 2*dr;
    c(N+1) = c_old(N+1) ...
           + Fo*(ghost - 2*c_old(N+1) + c_old(N)) ...
           + (Fo*dr/(2*r(N+1)))*(ghost - c_old(N));
end
% Plot result
figure; %[output:6f1e0723]
plot(r,c,'LineWidth',2); %[output:6f1e0723]
xlabel('r'); %[output:6f1e0723]
ylabel('c(r,t)'); %[output:6f1e0723]
title('Axisymmetric Cylindrical Diffusion'); %[output:6f1e0723]
grid on; %[output:6f1e0723]
```

Result:



Conclusion: The axisymmetric cylindrical diffusion equation was successfully solved using the FTCS finite-difference method. The singularity at (r=0) was handled using L'Hospital's rule, while the boundary condition at (r=1) was implemented using a ghost node. Stable numerical results were obtained for (N=50) and (Fo=0.24), and the concentration profiles showed the expected diffusion behavior over time.

Github Repository: https://github.com/mugdhajha/Project-Research